

Lecture 4

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Statistical Inference on Predictive and Causal Effects in High Dimensional Linear Regression Models

Introduction

- We recall the predictive effect question: How does the predicted value of Y change if a regressor D increases by a unit, while other regressors W remain unchanged?
- As before, we denote the set of regressors as $X = (D, W)$. In Lecture 1, we discussed how we could use the population regression coefficient corresponding to the variable D , denoted α , to answer this question.
- Now we turn to estimation and construction of confidence intervals for α in the high-dimensional setting.
- Here we focus on using Lasso methods first.

Causal interpretation of predictive effects

- The predictive effect question may have a causal interpretation within any model where conditioning on W is sufficient for identification of the structural/causal effect of D on Y .
- If conditioning on W is sufficient for identification of the causal effect of D on Y , the predictive effect question becomes a causal effect question: How does the predicted value of Y change if we **intervene** and increase D by a unit, holding W fixed?

Inference with Double Lasso

- The key to inference will be the application of Frisch-Waugh-Lovell partialling-out. Consider the simple predictive model:

$$Y = \alpha D + \beta' W + \epsilon,$$

where D is the target regressor and W consists of p controls. After partialling-out W ,

$$\tilde{Y} = \alpha \tilde{D} + \epsilon, \quad \mathbb{E} \epsilon \tilde{D} = 0,$$

where the variables with tildes are residuals retrieved from taking out the linear effect of W (practically, via linear regression)

$$\tilde{Y} = Y - \gamma'_{YW} W, \quad \gamma_{YW} = \arg \min_{\gamma \in \mathbb{R}^p} \mathbb{E}[(Y - \gamma' W)^2],$$

$$\tilde{D} = D - \gamma'_{DW} W, \quad \gamma_{DW} = \arg \min_{\gamma \in \mathbb{R}^p} \mathbb{E}[(D - \gamma' W)^2].$$

- The target parameter α can then be recovered from population linear regression of $\tilde{Y}$ on $\tilde{D}$:

$$\alpha = \arg \min_{a \in \mathbb{R}} \mathbb{E}[(\tilde{Y} - a\tilde{D})^2] = (\mathbb{E}[\tilde{D}^2])^{-1} \mathbb{E}[\tilde{D}\tilde{Y}].$$

- Note also that $a = \alpha$ solves the moment equation:

$$\mathbb{E}[(\tilde{Y} - a\tilde{D})\tilde{D}] = 0.$$

- We now consider estimation of α in a high-dimensional setting. For estimation purposes, we maintain that we have a random sample $(Y_i, X_i)_{i=1}^n$.
- To estimate α , we will mimic the partialling-out procedure in the population in the sample. In Lecture 1, where p/n was small, we employed ordinary least squares as the prediction method in the partialling-out steps. We are now considering cases where p/n is not small, and we instead employ Lasso-based methods in the partialling-out steps.

The Double Lasso

The estimation procedure for a target parameter α in a high-dimensional linear model setting can be summarized as follows:

1. We run Lasso regressions of Y_i on W_i and D_i on W_i :

$$\hat{\gamma}_{YW} = \arg \min_{\gamma \in \mathbb{R}^p} \sum_i (Y_i - \gamma' W_i)^2 + \lambda_1 \sum_j \hat{\psi}_j |\gamma_j|,$$

$$\hat{\gamma}_{DW} = \arg \min_{\gamma \in \mathbb{R}^p} \sum_i (D_i - \gamma' W_i)^2 + \lambda_2 \sum_j \hat{\psi}_j |\gamma_j|,$$

and obtain the resulting residuals:

$$\check{Y}_i = Y_i - \hat{\gamma}'_{YW} W_i,$$

$$\check{D}_i = D_i - \hat{\gamma}'_{DW} W_i.$$

In place of Lasso, we can use Post-Lasso or other Lasso relatives.

2. We run the least squares regression of $\check{Y}_i$ on $\check{D}_i$ to obtain the estimator $\check{\alpha}$:

$$\begin{aligned}\check{\alpha} &= \arg \min_{a \in \mathbb{R}} \mathbb{E}_n (\check{Y}_i - a \check{D}_i)^2 \\ &= (\mathbb{E}_n \check{D}_i^2)^{-1} \mathbb{E}_n \check{D}_i \check{Y}_i.\end{aligned}$$

- We can use standard results from this regression, ignoring that the input variables were previously estimated, to perform inference about the predictive effect, α .
- Good performance of the Double Lasso procedure relies on approximate sparsity of the population regression coefficients γ_{YW} and γ_{DW} .

Adaptive Inference with Double Lasso

Theorem 1 Under the stated approximate sparsity and additional regularity conditions, the estimation error in $\check{D}_i$ and $\check{Y}_i$ has no first order effect on $\check{\alpha}$, and

$$\sqrt{n}(\check{\alpha} - \alpha) \approx \sqrt{n} \mathbb{E}_n \tilde{D} \epsilon / \mathbb{E}_n \tilde{D}^2 \xrightarrow{D} N(0, V),$$

where

$$V = (\mathbb{E} \tilde{D}^2)^{-1} \mathbb{E}(\tilde{D}^2 \epsilon^2) (\mathbb{E} \tilde{D}^2)^{-1}.$$

- This statement means that $\check{\alpha}$ concentrates in a $\sqrt{V/n}$ -neighborhood of α , with deviations controlled by the normal law. Observe that the approximate behavior of the double lasso estimator is the same as the approximate behavior of the least squares estimator in low-dimensional models; see Lecture 1.
- Just like in the low-dimensional case, we can use these results to construct a confidence interval for α . The standard error of $\check{\alpha}$ is

$$\sqrt{\hat{V}/n},$$

where $\hat{V}$ is a plug-in estimator of V . The result implies, for example, that the interval

$$[\check{\alpha} \pm 2\sqrt{\hat{V}/n}]$$

covers α about 95% of the time.

Application to Testing the Convergence Hypothesis

- We provide an empirical example of partialling-out with Lasso to estimate the regression coefficient α in the high-dimensional linear regression model:

$$Y = \alpha D + \beta' W + \epsilon.$$

- In this example, we are interested in how economic growth rates (Y) are related to the initial wealth levels in each country (D) controlling for a country's institutional, educational, and other similar characteristics (W).
- The relationship is captured by α , the “speed of convergence/divergence”, which predicts the speed at which poor countries catch up ($\alpha < 0$) or fall behind ($\alpha > 0$) rich countries, after controlling for W .

- Our inference question here is do poor countries grow faster than rich countries, controlling for educational and other characteristics?
- In other words, is the speed of convergence negative: $\alpha < 0$? This is the Convergence Hypothesis predicted by the Solow growth model (Robert M. Solow is a world-renowned MIT economist who won the Nobel Prize in Economics in 1987).
- Under some **strong assumptions** that we won't state here, the predictive exercise we are doing can be given causal interpretation.

Data

- The outcome (Y) is the realized annual growth rate of a country's wealth (Gross Domestic Product per capita).
- The target regressor (D) is the initial level of the country's wealth.
- The controls (W) include measures of education levels, quality of institutions, trade openness, and political stability in the country.
- The sample contains 90 countries and about 60 controls.
- Thus $p \approx 60$, $n = 90$ and p/n is not small.
- We expect the least squares method to provide a poor/ noisy estimate of α . We expect the method based on partialling-out with Lasso to provide a high-quality estimate of α .

Results

	Estimate	Std. Error	95% - CI
OLS	-0.009	0.030	[-0.071, 0.052]
Double Lasso	-0.050	0.014	[-0.078, -0.022]

- Least squares provides a rather noisy estimate of the speed of convergence α which does not allow drawing strong conclusions about the convergence hypothesis.
- For example, the 95% confidence interval is very wide and includes both positive and negative values.
- In sharp contrast, Double Lasso provides a precise estimate. The lasso-based point estimate is -5% and the 95% confidence interval for the (annual) rate of convergence is -7.8% to -2.2% . This empirical evidence is consistent with the conditional convergence hypothesis.
- The corresponding notebook of the empirical analysis can be found [here](#).

Inference on Many Coefficients

- If we are interested in more than one coefficient, we can repeat the one-by-one Double Lasso procedure for each of the coefficients of interest and obtain valid estimation and inference on each component under regularity conditions.
- Here we consider the model

$$Y = \sum_{\ell=1}^{p_1} \alpha_{\ell} D_{\ell} + \sum_{j=1}^{p_2} \beta_j \bar{W}_j + \epsilon,$$

where the number of interesting predictors p_1 could be very large and the number of controls p_2 could also be very large.

Motivation of this model

There are at least three motivations for considering many coefficients of interest:

1. there can be multiple policies whose predictive effect we would like to infer
2. we can be interested in heterogeneous predictive effects across groups
3. we can be interested in nonlinear effects of policies

- The methodology can be used to study **heterogeneous effects**, where D'_ℓ s are generated as

$$D_\ell = D_0 \bar{X}_\ell, \quad \ell = 1, \dots, p_1,$$

where D_0 is the base variable of interest (a treatment indicator, price, group indicator), and $(\bar{X}_l)_{l=1}^{p_1}$ are known transformations of controls $\bar{W}$, for example various subgroup indicators.

- The methodology can also be used to study **nonlinear effects**. For example, we could consider D_ℓ 's generated as polynomial transformations of the base treatment variable:

$$D_\ell = D_0^\ell, \quad \ell = 1, \dots, p_1.$$

We could also further interact these transformations with other variables to study nonlinear heterogeneous effects.

One by One Double Lasso

- For each $\ell = 1, \dots, p_1$, we apply the one-by-one Double Lasso procedure for estimation and inference on the coefficient α_ℓ in the model

$$Y = \alpha_\ell D_\ell + \gamma'_\ell W_\ell + \epsilon, \quad W_\ell = ((D_k)'_{k \neq \ell}, \bar{W}')'.$$

- Under approximate sparsity conditions, the Double Lasso method provides a high-quality estimate $\hat{\alpha} = (\hat{\alpha}_\ell)_{\ell=1}^{p_1}$ of $\alpha = (\alpha_\ell)_{\ell=1}^{p_1}$ and we can construct individual confidence intervals, or even joint confidence bands.
- Under regularity conditions, this allows for simultaneous inference on $p_1 > n$ coefficients. **What does simultaneous inference mean?**

Double Lasso for Many Coefficients

Theorem 2 Under regularity conditions that impose approximate sparsity in all partialling out steps we have that uniformly in $\ell = 1, \dots, p_1$:

$$\sqrt{n}(\hat{\alpha}_\ell - \alpha_\ell) \approx (\mathbb{E}_n[\tilde{D}_\ell^2])^{-1} \sqrt{n} \mathbb{E}_n[\tilde{D}_\ell \epsilon],$$

where the approximation holds in the supremum distance. Further, if $(\log p_1)^5/n$ is small, we have that

$$\sqrt{n}(\hat{\alpha} - \alpha) \xrightarrow{D} N(0, V),$$

where

$$V_{\ell k} = (\mathbb{E}[\tilde{D}_\ell^2])^{-1} \mathbb{E}[\tilde{D}_\ell \tilde{D}_k \epsilon^2] (\mathbb{E}[\tilde{D}_k^2])^{-1}.$$

- The latter result allows the construction of simultaneous confidence bands on all target parameters α_ℓ 's of the form:

$$CR = \times_{\ell=1}^{p_1} \left[\hat{\alpha}_\ell \pm c \sqrt{\hat{V}_{\ell\ell}/n} \right],$$

with critical value c chosen so that

$$\Pr(\alpha \in CR) \approx 1 - \alpha$$

where $1 - \alpha$ denotes the confidence level.

Discovering Heterogeneity in the Wage Pay Gap Analysis

- We apply the method of the preceding section to analyze heterogeneity of the wage pay gap using CPS 2012 data.
- To explore heterogeneity, we interact the female indicator with group indicators capturing marital status, education groups, geographical regions, and a third degree polynomial in experience.
- The following tables provide estimated coefficients, standard errors, pointwise p-values and the simultaneous 90% confidence band for the heterogeneous effects.
- Among other things, we see that being married predict a larger drop in wages for women, controlling for all other characteristics.
- The notebook can be found [here](#).

Results

	Estimate.	Std. Error	p-value
female	-0.15	0.05	0.00
female:widowed	0.14	0.09	0.13
female:divorced	0.14	0.02	0.00
female:separated	0.02	0.05	0.66
female:nevermarried	0.19	0.02	0.00
female:hsdo8	0.03	0.12	0.82
female:hsd911	-0.12	0.05	0.02
female:hsg	-0.01	0.02	0.50
female:cg	0.01	0.02	0.58
female:ad	-0.03	0.02	0.16
female:mw	-0.00	0.02	0.96
female:so	-0.01	0.02	0.67
female:we	-0.00	0.02	0.84
female:exp1	0.00	0.01	0.53
female:exp2	-0.16	0.05	0.00
female:exp3	0.04	0.01	0.00

Simultaneous 90% Confidence Intervals

	5%	95%
female	-0.29	-0.02
female:widowed	-0.12	0.39
female:divorced	0.08	0.20
female:separated	-0.11	0.15
female:nevermarried	0.13	0.24
female:hsd08	-0.35	0.41
female:hsd911	-0.26	0.02
female:hsg	-0.06	0.04
female:cg	-0.04	0.06
female:ad	-0.09	0.03
female:mw	-0.05	0.05
female:so	-0.06	0.04
female:we	-0.06	0.05
female:exp1	-0.02	0.03
female:exp2	-0.28	-0.04
female:exp3	0.02	0.06

Why Partialling-out works: Neyman Orthogonality

- In the double lasso approach, α is the target parameter and η are *nuisance* parameters with true value

$$\eta^o = (\gamma'_{DW}, \gamma'_{YW})'.$$

- As the learned value of α depends on the values of the nuisance parameters, it is useful to explicitly consider the dependence of α on the nuisance parameters: $\alpha(\eta)$.
- The main idea of the double lasso approach is that it provides a procedure for learning the target parameter α that is first-order insensitive to local perturbations of the nuisance parameters around their true values, η^o :

$$D = \partial_{\eta} \alpha(\eta^o) = 0.$$

- We will call local insensitivity of target parameters to nuisance parameters *Neyman orthogonality*.

- Neyman orthogonality is important for providing high-quality estimation and inference, especially in high-dimensional settings.
- In high-dimensional settings, we use regularization procedures to estimate the nuisance parameters as solutions to suitable prediction problems.
- The use of regularization generally results in bias, and we may heuristically view using regularized estimates of nuisance parameters as plugging in estimates of these parameters that are close to, but not exactly equal to, the true values of the nuisance parameters η^o .
- Neyman orthogonality, which guarantees that the target parameter is locally insensitive to perturbations of the nuisance parameters around their true values, then ensures that this bias does not transmit to the estimation of the target parameter, at least to the first order.

Moment conditions

- The parameter of interest α that depends on nuisance parameters η with true value η^o is Neyman Orthogonal with respect to these parameters if

$$D = \partial_{\eta} \alpha(\eta^o) = 0.$$

- If the parameter α is defined as a root in a of the equation $M(a, \eta) = 0$, which depends on the nuisance parameters η with true value η^o , then the equation is Neyman orthogonal if

$$\partial_{\eta} M(\alpha, \eta^o) = 0.$$

- The principle is applicable to problems outside the high-dimensional linear model problem considered in this chapter.

What happens if we don't have Neyman Orthogonality?

- If we don't have Neyman orthogonality, we should not expect to get high-quality estimates of the target parameters.

Example 1 (Single Selection) In this invalid method, one applies Lasso regression of Y on D and W to select relevant covariates W_Y , in addition to the covariate of interest, then refits the model by least squares of Y on D and W_Y . Inference for the target parameter is then carried out using conventional inference based on the latter regression.

- Despite its simplicity and seeming intuitive appeal, the approach outlined above is not a valid approach if the goal is to perform inference on α . It is a fine approach if the goal is solely prediction of the outcome, but it can result in very misleading conclusions about the parameter of interest α , as we demonstrate below.

- The naive approach relies on the moment condition

$$\mathbf{M}(a, b) = \mathbb{E}[(Y - aD - b'X)D] = 0.$$

- This moment condition is satisfied by the true value, $a = \alpha$ when $b = \beta$, where it coincides with the classical moment condition for α underlying low-dimensional ordinary least squares which sets prediction errors to be orthogonal to each predictor variable.
- However, this moment condition does not exhibit Neyman Orthogonality since

$$\partial_b \mathbf{M}(\alpha, \beta) = \mathbb{E}[DX] \neq 0$$

unless D is orthogonal to X .

- Because this moment is not Neyman orthogonal, we would expect that the bias and the relatively slow rate $\sqrt{s \log(p \vee n)/n}$ of convergence of our estimate of $\beta'X$ would transmit to bias and slow convergence in estimates of α provided by solving the empirical analog of the naive moment condition.

Simulation Study

- We compare the performance of the naive and orthogonal methods in a computational experiment where $p = n = 100$ and $\beta_j = 1/j^2$, $(\gamma_{DW})_j = 1/j^2$, and

$$Y = 1 \cdot D + \beta'W + \varepsilon_Y, \quad W \sim N(0, I), \quad \varepsilon_Y \sim N(0, 1)$$
$$D = \gamma'_{DW}W + \tilde{D}, \quad \tilde{D} \sim N(0, 1)/4.$$

- From the histograms shown in the [Figure 1](#), we see that the naive estimator is heavily biased, as expected from the lack of Neyman orthogonality in its estimation strategy, while the orthogonal estimator, see [Figure 2](#), based on partialling out, is approximately unbiased and Gaussian.

Histograms

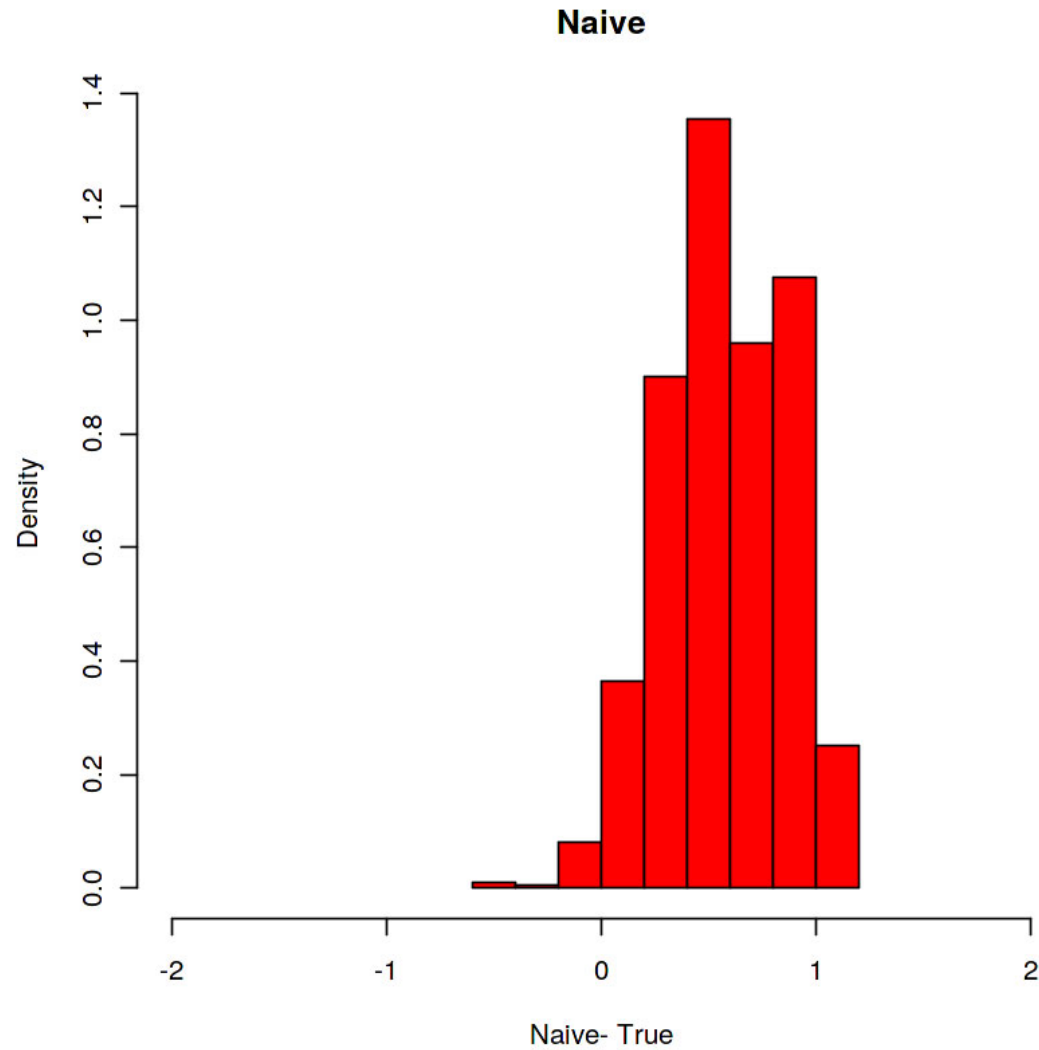


Figure 1: Naive Estimator (Simulated distribution of the naive estimator centered around the true value.)

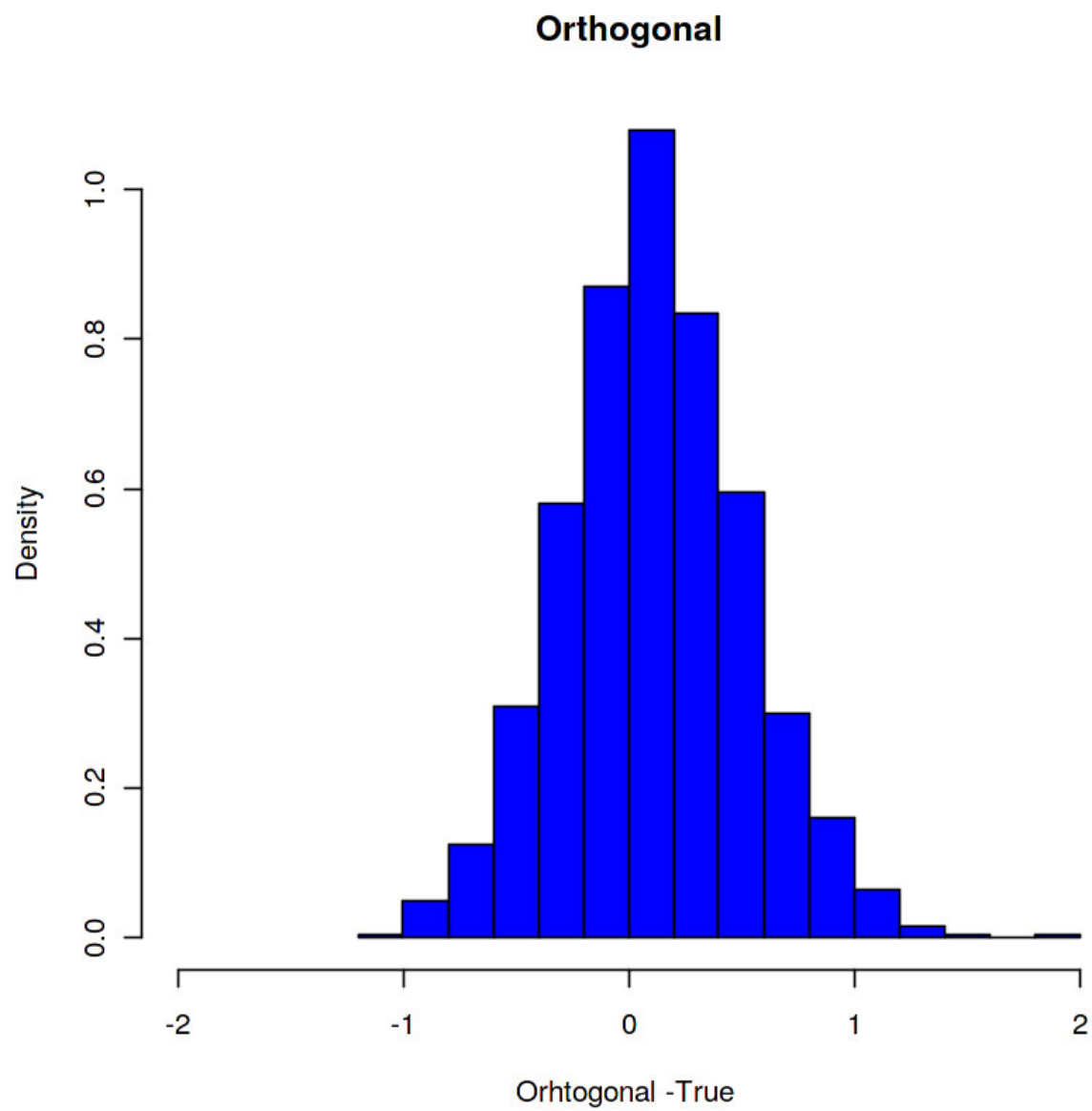


Figure 2: Double Lasso (Simulated distribution of the orthogonal estimator centered around the true value.)

- The reason that the naive estimator does not perform well is that it only selects controls W_j 's that are strong predictors of the outcome, thereby omitting weak predictors of the outcome. However, weak predictors of the outcome could still be strong predictors of D , in which case dropping these controls results in a strong **omitted variable bias**.
- In contrast, the orthogonal approach solves two prediction problems – one to predict Y and another to predict D – and finds controls that are relevant for either. The resulting residuals are therefore approximately “de-confounded”.
- An shiny app illustrating this problem can be found [here](#).

Double Selection

- One way to fix “single selection” would be to have “double selection”:
 1. find controls W_Y that predict Y as judged by lasso
 2. find controls W_D that predict D as judged by lasso
 3. regress Y on D and the union of controls $W_Y \cup W_D$; proceed with standard inference
- This procedure is approximately equivalent to the partialling out approach, and therefore inherits the orthogonality property.
- This approach is more conservative compared to single selection, as it makes sure that we have not omitted controls that are strong confounders for D . It therefore guards against large omitted variable biases.

Desparsified Lasso

- Yet another procedure that has the orthogonality property and is approximately equivalent to the partialling out approach under suitable conditions is *desparsified lasso*:
 1. Run a lasso estimator of Y on D and W , and save the coefficient estimate $\hat{\beta}$.
 2. Run a lasso estimator of D on W and save the coefficient estimate $\hat{\gamma}$.
 3. The estimator $\hat{\alpha}$ is then the solution of the empirical analog of the moment condition above:

$$\mathbb{E}_n[(Y - \hat{\alpha}D - \hat{\beta}'W)\tilde{D}(\hat{\gamma})] = 0,$$

where $\tilde{D}(\gamma) = D - \gamma'W$.

- Estimators of this form are referred to in econometrics as “instrumental variable estimators”. In purely technical terms we are using residualized $\tilde{D}$ to “instrument” for D .

Notebooks

- Notebook on the the Convergence Hypothesis.

CML